

Project 7

Instructions:

1. All questions must be answered using R.
2. A code must be written for each question in R. Save all codes in R-script.
For example, to answer question 1a) on the R script, first label the question number as #1a on the line before your code.
3. In addition to the code, a report (written in MS WORDS and **Converted to PDF format**) must be written explaining the results generated from your R code. The report must contain all outputs, files, and graphs generated from your code.
4. You are not allowed to discuss your project work with anyone except your professor and group members.
5. Your work must not be the same as that of your group members. This will be counted as exam malpractice and/or plagiarism with a huge penalty
6. Save your R file as Lastname_Firstname_GroupNumber
7. Submit both the R script, your typed MSWORD report, and every other generated files online on D2L.

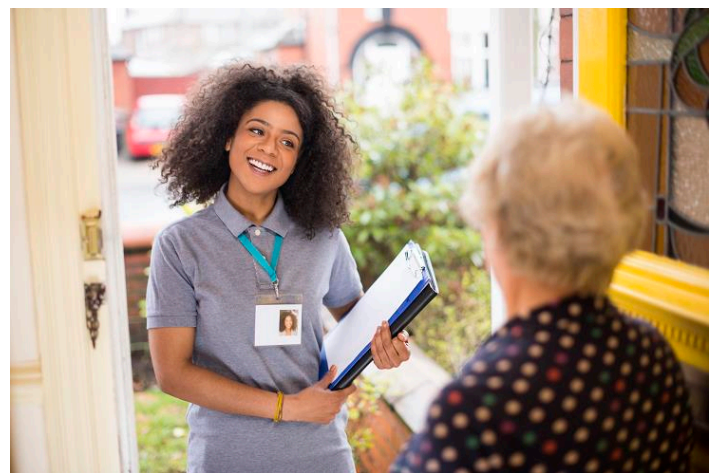
Chapter 7:

Background:

The US Census Bureau is a powerful agency in the United States that is responsible for providing data to the US government. It produces data which is used to allocate budget resources to different communities in the United States, and even determine how many Congressional representatives each state in the US is permitted. In order to create a census, the Census Bureau has to send out a large number of census workers across the United States, or send mail to households across the entire United States to get data on all US households.

A census worker greets someone at their front door.

The Census Bureau groups data on households into census tracts, where each census tract has a total population of about 4,000 residents. Census tracts should be divided so that the households in the census tracts share certain characteristics, such as economic status. Even though most tracts are around 4,000 residents, there is some variability, which is the focus of this lab. In practice, census tracts usually have a population between 1,200 and 8,000 people.



Guidelines

In this lab, you will do the following.

- Calculate the sample size for estimating the average population of the United States' census tracts to within a specified margin of error.
- Use the calculated sample size n in order to take simple random samples of the census tracts and calculate confidence intervals to demonstrate that you are consistently within a reasonable margin of error.

About the Data

For this lab, you need to download the [uscensuspopbytract.csv](#) dataset, which comes from the US Census.

- **POP100**—the total number of people living in the census tract at the time of the 2010 census

Tech Guide

You will need to use the [R Tech Guide](#).

Question:

1. Calculate the true mean μ and true standard deviation σ of the census tract populations based on all the rows in the data file, which has information about all of the census tracts in the United States. Use the formulas or correction factors in your software for the population mean and standard deviation. (Round your answer to two decimal places.)

$$\begin{aligned}\mu &= 4226.09 \\ \sigma &= 1981.86\end{aligned}$$

2. Each tract has its own population size, but all the tracts combined have an average population size μ , which you calculated above. The bureau wants the ability to create accurate estimates of the average population size μ of its census tracts in the future without having to conduct an expensive full census during a non-census year. Instead, they want to perform population surveys over a sample of the census tracts. They want to ensure their estimate is accurate to within 110 people of the true mean census tract population, with a confidence level of 95%. We will use the current data to make estimates of how large these future samples would need to be to achieve their objectives.

(a) What z_c critical value from the normal distribution corresponds to a 95% confidence interval? (Round your answer to two decimal places.) **1.96**

(b) Let E is the specified maximal margin of error. Calculate the sample size n for estimating μ when σ is known. (Round your answer up to the nearest whole number.) **1248**

4. In order to convince the bureau that the n you calculated would achieve the right level of accuracy, with the specified level of confidence, create 10 samples, and use the samples to create estimates of the population mean. For each of these, create confidence intervals using the following.

Sample	Interval
Sample 1	$\underline{4141.35} < \mu_1 < \underline{4361.27}$
Sample 2	$\underline{4049.53} < \mu_2 < \underline{4269.45}$
Sample 3	$\underline{4143.63} < \mu_3 < \underline{4363.55}$
Sample 4	$\underline{4059.85} < \mu_4 < \underline{4279.77}$
Sample 5	$\underline{4054.07} < \mu_5 < \underline{4273.99}$
Sample 6	$\underline{4096.73} < \mu_6 < \underline{4316.64}$
Sample 7	$\underline{4201.77} < \mu_7 < \underline{4421.69}$
Sample 8	$\underline{4144.91} < \mu_8 < \underline{4364.82}$
Sample 9	$\underline{4164.22} < \mu_9 < \underline{4384.13}$
Sample 10	$\underline{4044.63} < \mu_{10} < \underline{4264.54}$

5. About 95% of your confidence intervals should contain the true mean μ , which you calculated above. Is this true? Discuss.

All 10 of the intervals in the random seed I picked contain μ .
 To test further, I wrote another loop that checks against 30,000 samples created the same way, which had a 94.9033% success of μ being within the interval. 3000 should be sufficiently large to test.